

Title: Estimating efficacy over time of RSV prevention strategies in children and its impact on cost-effectiveness.

Research context: Respiratory Syncytial Virus (RSV) causes substantial morbidity and mortality, especially in very young children and the elderly. Recently, preventive products have been developed and approved, including maternal antibodies for newborns at birth (mAb) and maternal vaccination (MV). Efficacy of these products has been measured during clinical trials, but the data on how these products efficacy wanes over time are scarce. It is however crucial for health care decision making to understand the effectiveness of these products over time as well as their cost-effectiveness.

Research questions:

- (1) How does the efficacy of RSV mAb and MV changes over time?
- (2) How does the efficacy of RSV mAb and MV impact on the cost-effectiveness of these products for preventing RSV in children?

Data:

Published data are available from Kampmann et al 2023. The cost-effectiveness model evaluating cost-effectiveness of mAb and MV in preventing RSV in children is available within the research group CHERMID and is implemented in the software R.

Expected contribution of the student:

The student is expected to apply different statistical models to the available efficacy data to estimate efficacy over time. As part of the project, the student will keep up to date with the literature on the assumptions and models that are used to estimate RSV mAb and MV over time. If time allows, the student can then implement the different efficacy estimates into the cost-effectiveness model to evaluate their impact on the cost-effectiveness.

Supervisors:

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